Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering & Information Systems

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Programme : M.Tech (SE)
Course Title : Agile Development Process and Devops
Slot : C2+ TC2
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Question:

2. Order Tracking System
Problem Statement:
Ensure order status updates are correctly processed before release.
Tasks:
· Develop order lifecycle logic
· Write JUnit tests for status transitions
· CI/CD pipeline for automated deployment
· Deploy on Kubernetes
Real-world relevance: Flipkart tracking

<u>Implementation:</u>

<u>App.java</u>

```
package com.example;
import java.util.*;

public class App {

    // Order class
    static class Order {
        int id;
        String status;

        Order(int id) {
            this.id = id;
            this.status = "CREATED";
        }
    }

    static List<Order> orders = new ArrayList<>();

    // CREATE ORDER
    public static void createOrder(int id) {
        orders.add(new Order(id));
    }

    // UPDATE STATUS (Lifecycle logic)
    public static void updateStatus(int id, String newStatus) {
        for (Order o : orders) {
            if (o.id == id) {

                // Allowed transitions
                if (o.status.equals("CREATED") &&
newStatus.equals("SHIPPED")) {
                    o.status = newStatus;
                } else if (o.status.equals("SHIPPED") &&
newStatus.equals("DELIVERED")) {
                    o.status = newStatus;
```

```
                } else {
                    System.out.println("Invalid Transition!");
                }
            }
        }
    }

    // READ STATUS
    public static String getStatus(int id) {
        for (Order o : orders) {
            if (o.id == id) return o.status;
        }
        return "NOT FOUND";
    }

    public static void main(String[] args) {

        createOrder(1);

        System.out.println(getStatus(1)); // CREATED

        updateStatus(1, "SHIPPED");
        System.out.println(getStatus(1)); // SHIPPED

        updateStatus(1, "DELIVERED");
        System.out.println(getStatus(1)); // DELIVERED
    }
}
```

AppTest.java

```
package com.example;
import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testValidTransition() {
        App.orders.clear();

        App.createOrder(1);
        App.updateStatus(1, "SHIPPED");

        assertEquals("SHIPPED", App.getStatus(1));
    }

    @Test
    public void testInvalidTransition() {
        App.orders.clear();
```

```java
        App.createOrder(1);
        App.updateStatus(1, "DELIVERED"); // invalid

        assertEquals("CREATED", App.getStatus(1));
    }

    @Test
    public void testFinalState() {
        App.orders.clear();

        App.createOrder(1);
        App.updateStatus(1, "SHIPPED");
        App.updateStatus(1, "DELIVERED");

        assertEquals("DELIVERED", App.getStatus(1));
    }
}
```

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>order-tracking</artifactId>
    <version>1.0</version>

    <properties>
        <java.version>17</java.version>
    </properties>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>


            <plugin>
```

```
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
                <configuration>
                    <release>17</release>
                </configuration>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.3.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>

        </plugins>
    </build>

</project>
```

Pipeline:

```
pipeline {
    agent any

    tools {
        maven 'M3'
    }

    stages {

        stage('Checkout Git') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/aishwaryasuresh020-
source/Order_tracking_system.git'
            }
        }

        stage('Build and Test') {
            steps {
                sh 'mvn clean test'
            }
        }

    }
```

```
}
```

Dockerfile

```
FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/order-tracking-1.0.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

Deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: order-app-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: order-app
  template:
    metadata:
      labels:
        app: order-app
    spec:
      containers:
        - name: order-app
          image: order-app:latest
          imagePullPolicy: Never   # IMPORTANT for local Docker
          ports:
            - containerPort: 8080
```

Screenshots:

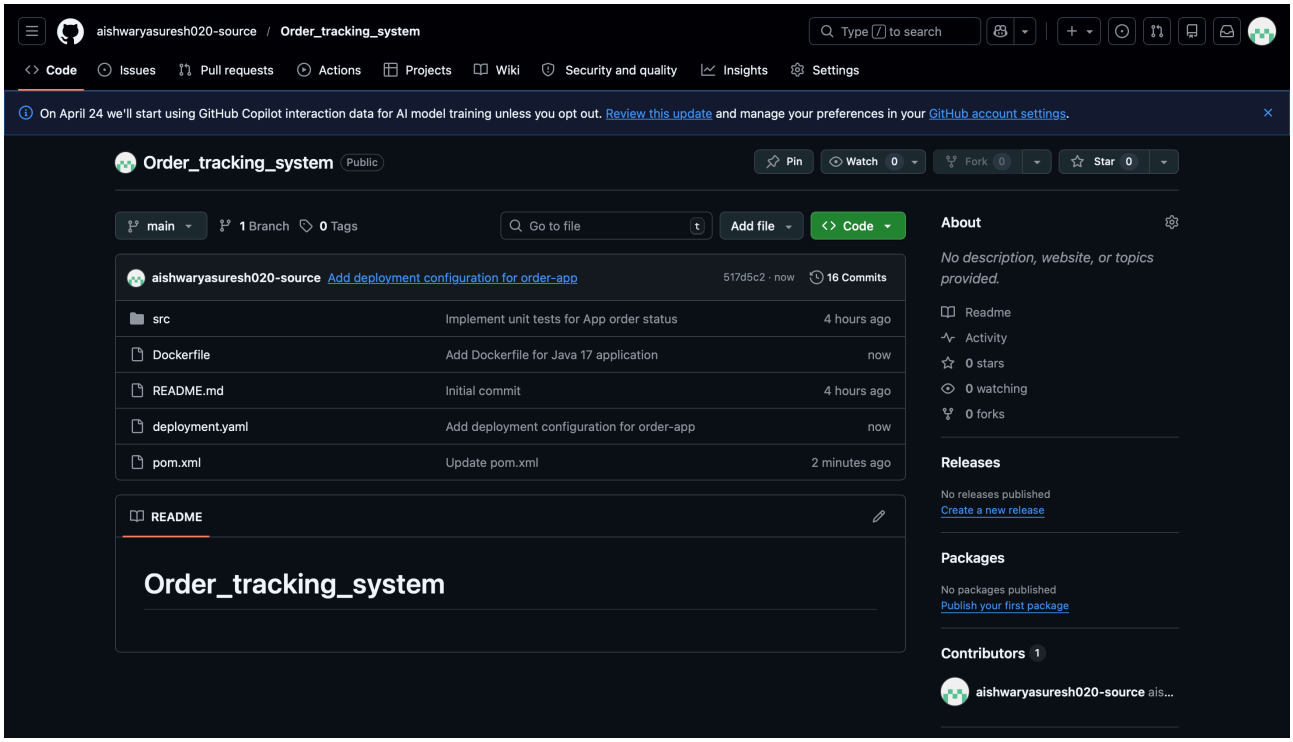
Git initialisation:

```
● aishwaryasuresh@Aishwaryas-MacBook-Air-2 Ots % git init
  Initialized empty Git repository in /Users/aishwaryasuresh/Downloads/Ots/.git/
● aishwaryasuresh@Aishwaryas-MacBook-Air-2 Ots % git clone https://github.com/aishwaryasuresh020-source/Order_tracking_system.git
  Cloning into 'Order_tracking_system'...
  remote: Enumerating objects: 25, done.
  remote: Counting objects: 100% (25/25), done.
  remote: Compressing objects: 100% (14/14), done.
  remote: Total 25 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
  Receiving objects: 100% (25/25), 6.15 KiB | 3.07 MiB/s, done.
  Resolving deltas: 100% (3/3), done.
```

JUnit test:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % mvn clean test
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
Invalid Transition!
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s - in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.541 s
[INFO] Finished at: 2026-04-12T17:12:19+05:30
[INFO] ------------------------------------------------------------------------
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system %
```

Github:



Jenkins CI/CD pipeline for automated deployment:

```
 > git config remote.origin.url https://github.com/aishwaryasuresh020-source/Order_tracking_system.git # timeout=10
 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision add31e40857ea07002a597ea7621fe8d4f82c71e (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f add31e40857ea07002a597ea7621fe8d4f82c71e # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git checkout -b main add31e40857ea07002a597ea7621fe8d4f82c71e # timeout=10
Commit message: "Update pom.xml"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build and Test)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:order-tracking >----------------------
[INFO] Building order-tracking 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ order-tracking ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ order-tracking ---
```

```
    --release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules
automatically
[INFO]
[INFO] --- surefire:3.0.0-M7:test (default-test) @ order-tracking ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
Invalid Transition!
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.017 s - in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.716 s
[INFO] Finished at: 2026-04-12T18:29:58+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Mvn clean package:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % mvn clean package
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
Invalid Transition!
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ order-tracking ---
[INFO] Building jar: /Users/aishwaryasuresh/Downloads/Ots/Order_tracking_system/target/order-tracking-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.797 s
[INFO] Finished at: 2026-04-12T21:48:43+05:30
[INFO] ------------------------------------------------------------------------
```

Docker implementation:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % docker rmi -f order-app
docker build -t order-app .
Error response from daemon: No such image: order-app:latest
[+] Building 2.1s (9/9) FINISHED                                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                                  0.0s
 => => transferring dockerfile: 155B                                                                                                  0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                                            2.0s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                               0.0s
 => [internal] load .dockerignore                                                                                                     0.0s
 => => transferring context: 2B                                                                                                       0.0s
 => CACHED [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3         0.0s
 => [internal] load build context                                                                                                     0.0s
 => => transferring context: 3.63kB                                                                                                   0.0s
 => [2/3] WORKDIR /app                                                                                                                0.0s
 => [3/3] COPY target/order-tracking-1.0.jar app.jar                                                                                  0.0s
 => exporting to image                                                                                                                0.0s
 => => exporting layers                                                                                                               0.0s
 => => exporting manifest sha256:f1e63597bf39a5684ae9a3769e889c40976b16c77e4e6406fac2e6c22899c99f                                     0.0s
 => => exporting config sha256:28c0b100186246796d508b6abe09a9c8e3f684a5e9f3877e05a7b2810c9a0311                                       0.0s
 => => exporting attestation manifest sha256:38768519faffb1a4c2c6e1d1c1607fcdb3d18a92325ba1ee1c88327713602987                         0.0s
 => => exporting manifest list sha256:9e620d160d548043b3139dc4fe48e75fe01c7a1d4991a075024b9473aea8d3a2                                0.0s
 => => naming to docker.io/library/order-app:latest                                                                                   0.0s
 => => unpacking to docker.io/library/order-app:latest                                                                                0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/1tcmqakr3zo0gj2r5pebo9rsf
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % docker tag order-app aishwaryasuresh/order-app:latest
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % docker login
docker push aishwaryasuresj/order-app:latest
Authenticating with existing credentials... [Username: aishwaryasuresj]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
The push refers to repository [docker.io/aishwaryasuresj/order-app]
tag does not exist: aishwaryasuresj/order-app:latest
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % kubectl apply -f deployment.yaml
deployment.apps/order-app-deployment created
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system % kubectl get pods
NAME                                  READY   STATUS             RESTARTS       AGE
order-app-deployment-74bdc75d97-zf97d   0/1     ErrImageNeverPull   0              75s
resume-deployment-84c4cc5dfc-kqzcz      1/1     Running             6 (112m ago)   7d4h
resume-deployment-84c4cc5dfc-xgz9r      1/1     Running             6 (112m ago)   7d4h
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Order_tracking_system %
```

Docker Image:

Builds / Order_tracking_system

Order_tracking_system

1tcmqakr3zo0gj2r5pebo9rsf

Status	Duration	Builder	Author
Completed	2.0s	desktop-linux	N/A

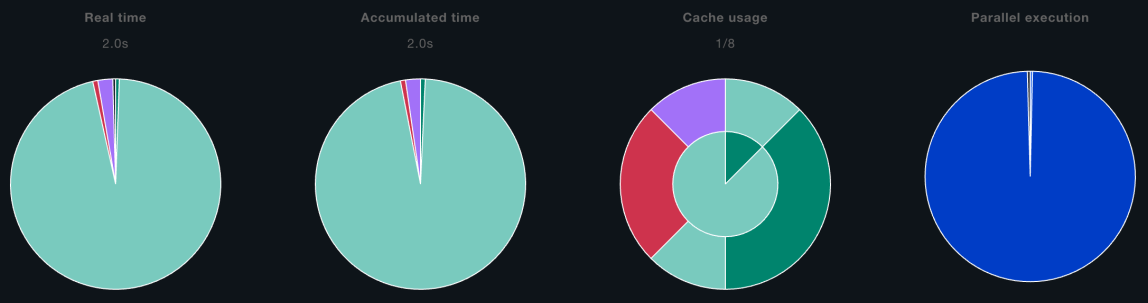
Info | Source | Logs | History

Source details

Filter by platform: linux/arm64

File name	Dockerfile
Remote source location	https://github.com/aishwaryasuresh020-source/Order_tracking_system.git
Revision	517d5c2

Build timing



Local file transfers | Image pulls | File operations | Result exports | Idle

Build start time	4/12/2026, 9:52:41 PM	Cached steps	1
Build end time	4/12/2026, 9:52:43 PM	Non-cached steps	7

Builds / Order_tracking_system

Order_tracking_system

1tcmqakr3zo0gj2r5pebo9rsf

Status	Duration	Builder	Author
Completed	2.0s	desktop-linux	N/A

Info | Source | Logs | History

Dependencies

Source	Digest
eclipse-temu	2ff0b91aaa4cb5dc35b476...

Order_tracking_system
- src
  - main/java/com/example
    - App.java
  - test/java/com/example
    - AppTest.java
- target
- deployment.yaml
- Dockerfile
- pom.xml
- README.md

Configuration

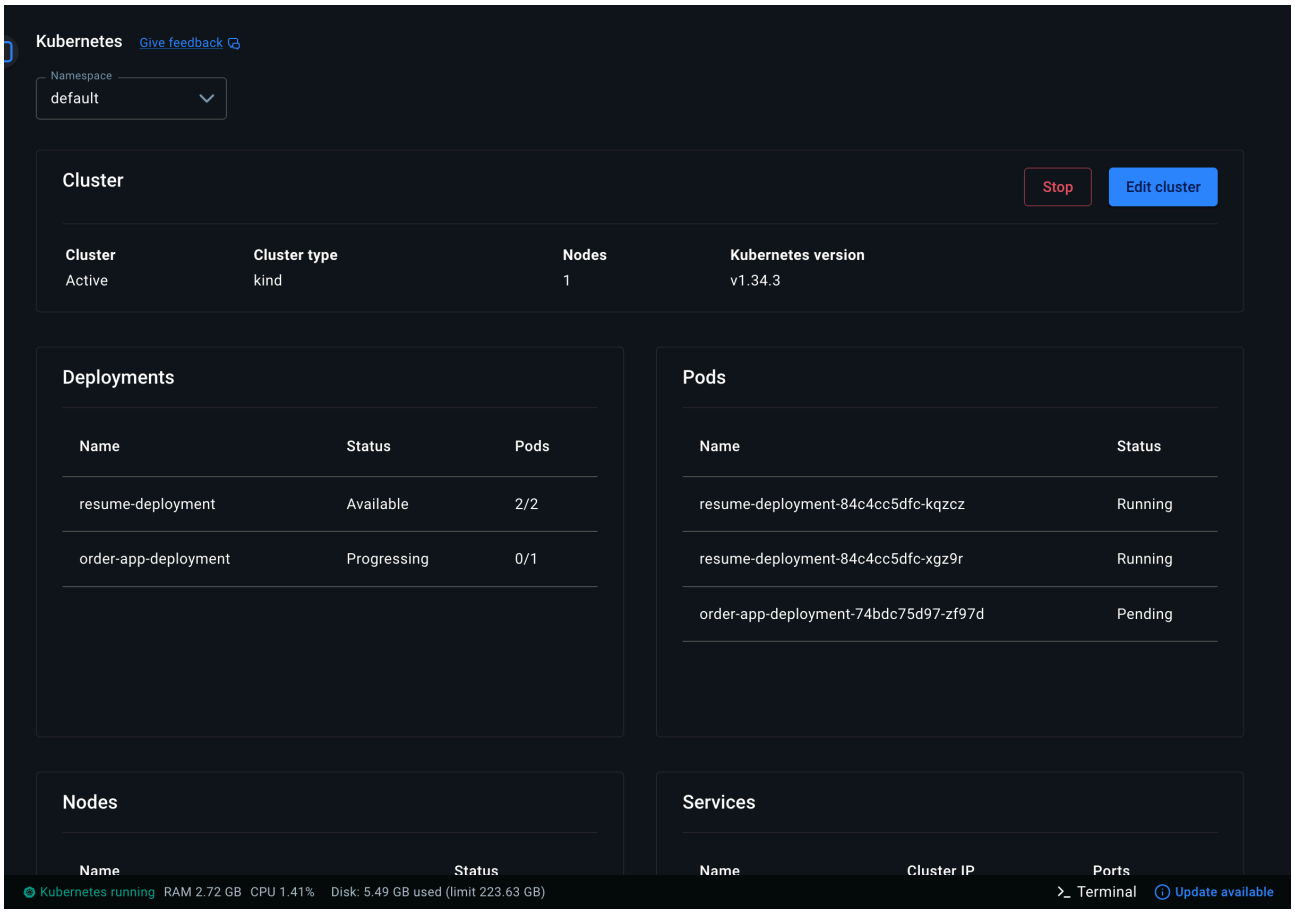
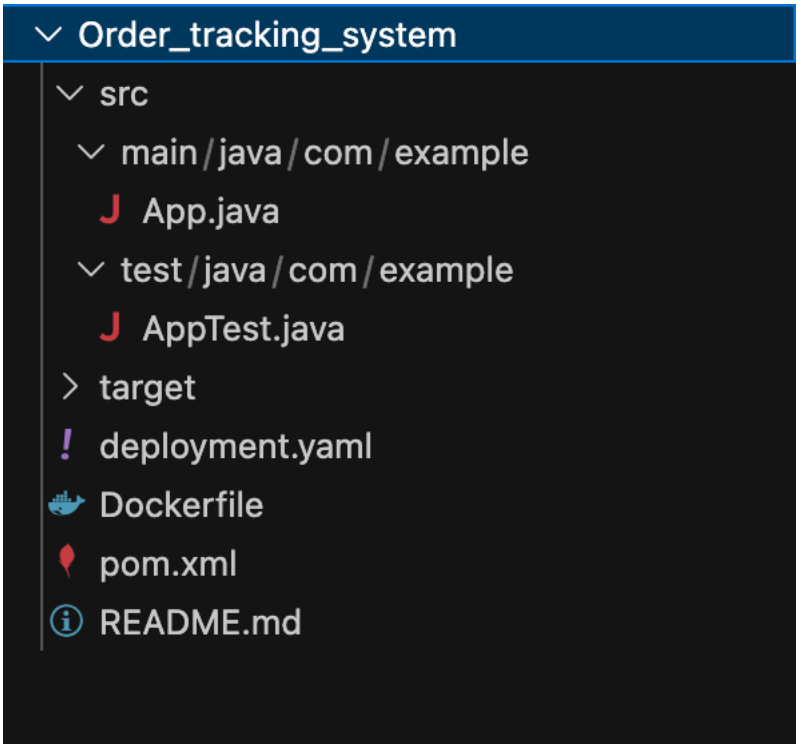
No configurations found

Build results

Artifact		Digest	Size
application/vnd.oci.i		9e620d160d5480...	856 Bytes
OpenTelemetry trace		3d9794e4e56f8c...	101.3 kB
Provenance v1	arm64	90c2b50ada423f...	16 kB

Tags	Digest
docker.io/library/order-app:latest	9e620d160d548043b3139dc4...

Kubernetes:



File structure: